CHAPTER 2
RELATED WORK

This section places the system within the search-and-rescue (SAR) robotics literature along the four challenges of Section 1.2: a bounded stop time (C1), freedom from single points of failure (C2), two-way victim interaction under constrained resources (C3) and affordability (C4). Sections 2.1 to 2.5 cover prior reviews, field deployments and platforms, fail-safe architectures, human-robot interaction (HRI) and real-time media; Section 2.6 states the gap and compares platforms in Table 2.1.

The search was not systematic and no PRISMA flow is claimed. In October 2026 OpenAlex returned 446 articles and conference papers from 2022 onward for "search and rescue robot teleoperation low-cost", 272 for "rescue robot victim communication two-way audio", 56 for "WebRTC robot teleoperation", 1 for "fail-safe watchdog dead-man teleoperated mobile robot" and 1,204 for "rescue robot Raspberry Pi"; these counts show how large the surrounding literature is, not how many records were screened. A second round looked for adjacent evidence: low-cost Raspberry Pi and Robot Operating System (ROS) rescue robots ("teleoperated search and rescue robot Raspberry Pi low-cost", 34 records; "ROS-based rescue robot teleoperation low-cost ground vehicle disaster", 61), link-loss failsafes ("communication loss failsafe teleoperated mobile robot", 11; "UAV failsafe lost link return to launch flight controller", 14), watchdog timers ("watchdog timer mobile robot safety stop", 65) and fault tolerance ("fault tolerance mobile robot software failure recovery supervisor", 970). Candidates were selected from the relevance-ranked results by title and abstract and by backward citation from the reviews of Section 2.1. Table 2.1 includes peer-reviewed platforms published from 2022 onward that present a built or simulated SAR robot and whose full text was read, with one earlier exception stated in Section 2.6. No study of a dead-man or command-timeout stop for a teleoperated rescue robot was found among the results screened.

2.1 Prior Reviews and Levels of Operator Involvement

Reviews of SAR robotics organise the field along different axes: a domain theory of robot-assisted search drawn from eight years of HRI experience [**?**], the control perspective [**?**], multi-robot coordination and perception [**?**], and readiness for deployment, where Delmerico et al. observe that field-deployable rescue technology has requirements for robustness, speed, versatility and ease of use that the state of the art may not meet [**?**]. Schneider and Wildermuth set out SAR requirements on each component of a robot [**?**]. Recent reviews organise the field by system architecture, from 74 papers of 2016-2024 [**?**]; by platform class, from more than 190

studies of unmanned ground vehicles (UGVs) of 2020-2025 [**?**]; by perception deficiencies [**?**]; by ethical and design issues [**?**]; and, for mines, by design requirements, finding that two-way voice is a fundamental functionality of most mine rescue robots [**?**]. The open problems they name include multi-robot coordination, real-time decision-making, energy and standardised benchmarks [**?**, **?**]; neither the timing of the stop path nor the cost of victim interaction on the robot is the axis along which they organise the field.

Levels of robot autonomy range from teleoperation to fully autonomous systems [**?**], and SAR robots are in practice either tele-operated or semi-autonomous [**?**]. This paper distinguishes Teleoperation (the operator drives the robot directly), Supervisory Control (the operator sets goals that the robot executes), Shared-Collaborative Control (operator and robot share decision authority) and Full Autonomy (no human in the control loop during the mission). The system presented here is a Teleoperation system, because rescue robots are typically teleoperated in actual disasters [**?**] and teleoperation is expected to remain an important default mode even for autonomous systems [**?**]. In the Urban Circuit of the DARPA Subterranean Challenge, where communication constraints make underground search harder, one team used teleoperation as the baseline against which autonomous solutions were compared [**?**]. Higher levels, which would combine the complementary capabilities of operators and assistance functions [**?**], are treated in Section 8 as extensions subordinate to the stop guarantee of Section 3.5.

2.2 Field Deployments, Benchmarks and Recent Platforms

Field deployments. The response to the World Trade Center collapse in September 2001 is the first known use of robots for urban search and rescue [**?**]; its post-hoc analysis, the first study of HRI during an unstaged rescue, called for perceptual and assistive interfaces and formal models of the robot's state [**?**]. At Fukushima Daiichi in 2011, tests in a comparable plant indicated that robust wireless communication would not be possible inside the reactor buildings, so the robot deployed there was fitted with a wired communication device [**?**]; organisation, operation scheme and systemisation were the major lessons [**?**]. After the 2016 Amatrice earthquake, ground and aerial robots collected data for 3D models of two damaged churches [**?**]. Because the Fukushima deployment had to be designed around the radio link failing, this paper treats link loss as an expected operating condition rather than a rare fault (C1, C2).

Benchmarks. Standard test methods for response robots have been refined through competitions [**?**], and the DARPA Subterranean Challenge (2018-2021) evaluated integrated systems in physical and simulated underground contests [**?**]. Section 5 does not use the standardised apparatus, so its results are not comparable with benchmark scores.

Recent platforms. The platforms in Table 2.1 concentrate their contribution on mobility, detection or communication range rather than on the control path. They range from a tracked robot with flippers and a manipulator scored at the World Robot Summit 2020 [**?**] to commodity designs: TXRob streams camera, microphone and gas data from a Raspberry Pi 4 over

local Wi-Fi [**?**], and Pramono et al. combine ESP32 controllers, Bluetooth control and Internet-connected video [**?**]. A quadruped built on ROS and a Raspberry Pi 4B, operated with a joystick, sends camera and depth-camera data over Wi-Fi and reports a 200 m control range [**?**]. Others extend range with a LoRa network that needs no infrastructure [**?**], use a LoRa command link with Wi-Fi video and a cloud dashboard [**?**], add voice control and person tracking to existing robots through a cloud-connected controller [**?**], or change the mobility class, as in a caged ground-air UAV [**?**] and an active scope camera with 80 distributed sensors [**?**]. Detection-oriented work locates survivors with a thermal infrared array [**?**], a YOLOv8 detector on a Jetson Nano [**?**], multimodal CNN-LSTM fusion validated in simulation [**?**], or Bluetooth Low Energy wearables detected up to 15 m away at smartphone-level transmit power [**?**]. Affordability is claimed more often than quantified: five of these designs are described as low-cost, affordable, cost-effective or built from inexpensive parts [**?**, **?**, **?**, **?**, **?**], yet none of the recent platforms in Table 2.1 states a numeric build cost; only the earlier Survivor Buddy head does, at USD 3,600 for version 1.0 and USD 1,800 for version 2.0 [**?**], so Section 5.9 reports this robot's figure as an estimate (C4).

2.3 Fail-Safe and Fault-Tolerant Architectures for Low-Cost Mobile Robots

Bounded stop. IEC 60204-1 defines stop categories for machinery, of which category 0 is stopping by immediate removal of power to the machine actuators, an uncontrolled stop [**?**]. The dead-man principle is not new; for a low-cost robot the questions are where the stop decision runs and whether its timing can be bounded, since a decision taken by a Linux process inherits its scheduling and crash behaviour. Industrial practice is moving safety functions from physical hardware such as safety relays and interlock circuits into software [**?**], which makes the placement of the stop decision a design choice. Pont presents patterns for time-triggered embedded systems, in which tasks are released by a periodic timer so that their timing can be predicted [**?**]; the firmware of Section 3.5 follows this style so that the stop does not depend on Linux (C1). It behaves like a category-0 stop but is not a safety-rated implementation in the sense of the standard. Watchdog timers are the usual mechanism for detecting a stalled controller: a hardware watchdog designed for drone processors resets the processor when it is not serviced within an interval [**?**]. That mechanism detects a hung processor, not the absence of operator commands. For loss of the operator link, the documentation of the open-source ArduPilot Rover autopilot describes a radio failsafe that triggers when the transmitter link is lost for a configurable timeout (default 1 s) and a ground-station failsafe that triggers when no heartbeat arrives within a configurable timeout (default 5 s), with actions that include holding position [**?**]. This documents configurable timeouts for general-purpose vehicle software rather than measured stop times on a rescue robot. A run-time monitor for networked robots under delayed observations [**?**] addresses

late information; a stop on missing commands covers commands that stop arriving, so the two are complementary.

Process supervision. Candea et al. argue that systems built to recover fast may be more productive than systems built never to fail, with multiple lines of defence because recovery can itself fail [**?**]. Schrick et al. use microservices and a hierarchical state machine to increase the interruptibility and fail-safety of a delivery robot [**?**], Ramesh et al. estimate "robot health" at run time so that a supervisor can intervene [**?**], and a review of humanoid-robot safety finds cybersecurity and standardisation addressed less often than physical safety [**?**]. At design time, failure mode and effects analysis combined with simulation has been used to assess how embedded software detects and mitigates random sensor and actuator hardware failures of a rover's mobility system [**?**]; that work addresses hardware faults, not the failure of supervised software processes. None of these works targets a single-board rescue robot. Section 3.7 applies recovery-oriented supervision to a rescue robot on a single-board computer (C2). No platform in Table 2.1 reports a hardware-level stop guarantee or supervised process recovery.

2.4 Human-Robot Interaction: Operator and Victim

Operator side. Situation awareness, modelled by Endsley through its role in dynamic decision making [**?**], is the recurring operator-side concern. Chen et al. summarise the performance decrements that limited video bandwidth, time lags, low frame rates and two-dimensional views cause in teleoperation [**?**]; a survey of vehicle teleoperation likewise identifies network latency as a critical factor, through its effect on operators' real-time perception and decision-making [**?**]; and Luz et al., for planetary exploration, report that even small latencies can compromise robot safety and raise workload [**?**]. For mine rescue, interface design principles have been mapped from cognitive load, situational awareness and trust [**?**], and rescue personnel report a need for real-time situational data and customisable interfaces [**?**]. Semantic indicators can signal when a scene needs the expert operator's attention [**?**], and LED-based affective expressions improved participants' situational awareness in an online study [**?**]. The dashboard of Section 3.6 addresses this operator-side concern.

Victim side. A SAR robot should be able to provide audio-visual communication between the operator and trapped persons that allows basic diagnosis by an emergency responder [**?**], and on average four to ten hours pass between discovery and extrication, during which the robot must keep the victim calm [**?**]. Bethel and Murphy applied non-verbal affective expression to victim assessment within about three metres of a victim [**?**]. Survivor Buddy, a four-degree-of-freedom head with a touch-screen face, was built to study robots as social media between trapped survivors and the outside world [**?**]; its second version was 50% less expensive and 78% lighter [**?**]. Its reported evaluations used a quasi-experimental study with four middle-school girls and their friends, which the authors describe as neither statistically significant nor rigorously controlled [**?**], and two simulated urban search and rescue scenario studies [**?**]. Two-way voice

is also a fundamental functionality of most mine rescue robots, such as the ANDROS Wolverine V2 [**?**], which are specialised platforms outside the low-cost scope of Table 2.1.
Among recent platforms the trapped person is mainly an object of detection [**?**, **?**, **?**, **?**, **?**, **?**]; one cloud-connected design examines the robot's video and audio to detect survivors and streams their location to a server [**?**]. Where speech toward the person is mentioned, it is described rather than tested: the Universal Platform's speech synthesis is said to be able to give protocol information to rescued individuals [**?**], and the speakers of the active scope camera are stated as used to talk to survivors [**?**], but the reported experiments of both test other functions. TXRob's microphone relays sound to the operator only [**?**]; SERR provisions a speaker and microphone for victims but has not yet implemented them [**?**]; and Rescuer gives the operator a push-to-talk walkie-talkie for audio with the victim, while its camera video is recorded on the robot rather than streamed [**?**]. Two-way victim communication is therefore an established requirement and a feature of fielded mine-rescue robots; what no recent low-cost platform in Table 2.1 demonstrates is two-way audio and video with the victim on single-board hardware, or its CPU, power and bandwidth cost (C3).

2.5 Real-Time Media and Computing on Edge Hardware

WebRTC has been used for teleoperation with strict latency targets: a remote dental surgery robot on Amazon Kinesis Video Streams WebRTC reports end-to-end latency below 50 ms, 34.2% lower than its baseline [**?**], and a multi-operator platform for a mobile manipulator, validated over the Internet, uses dedicated web and signalling servers and Google's public STUN servers [**?**]; Sections 3.8 and 3.9.4 address the same multi-operator problem with arbitration and floor control. Both depend on servers outside the robot's network, and TXRob streams one-way video over local Wi-Fi without WebRTC [**?**]; the robot described here runs both WebRTC endpoints and signalling on the local network (Sections 3.2, 3.9). Opus supports frame sizes from 2.5 ms to 60 ms, and the specification describes 20 ms as a good choice for most applications [**?**]. Bachhuber and Steinbach present a glass-to-glass delay measurement system with 0.5 ms precision [**?**]. Sections 3.9.5 and 5.2 return to the frame size and the delay metric.
The cloud-to-edge-to-IoT continuum has been proposed to meet the low-delay requirements of SAR, though building it remains an open challenge [**?**]. Recent platforms send alerts or streams through the cloud [**?**, **?**], presuming infrastructure that a collapsed site may not provide, the concern behind infrastructure-free LoRa [**?**] and the wired link at Fukushima [**?**]. Running control, two-way media and supervision together on a single-board computer without the cloud is not characterised for a rescue robot in these works; Section 5.5 reports that budget.

2.6 Research Gap

Table 2.1 compares representative rescue platforms against C1-C4. Field studies, benchmarks and surveys [**?**, **?**, **?**, **?**, **?**, **?**, **?**, **?**, **?**, **?**, **?**, **?**] are not tabulated. Survivor Buddy is included although it predates 2022 as the most developed academic work on the robot as a communication medium for the victim; fielded mine-rescue robots with two-way voice are reviewed elsewhere [**?**] and fall outside the table's low-cost, post-2022 scope.

Table 2.1. Representative rescue platforms against challenges C1-C4. Cells for other platforms are taken from the cited papers and not re-measured. "NR" means not reported in the text read; it does not mean the feature is absent. No cited paper states that it lacks a bounded stop (C1) or process supervision (C2), so every C1 and C2 cell for another platform is NR. "(stated)" marks a cell in which the paper itself says what is or is not provided. "Evidence and conditions" gives the kind of evidence and the test setting, which differ across rows (physical prototype, competition, simulation, dataset), so the rows are not comparable as performance results. "Text read" is the source read for the row. The table is split into two parts, Table 2.1a and Table 2.1b, with rows in the same order in both.

Table 2.1: Representative rescue platforms: platform, control and link/computing against C1-C2 (part a of Table 2.1).

Platform	**Platform and control**	**Link and computing**	**C1**	**C2**
Survivor Buddy 1.0/2.0 [**?**, **?**]	4-DoF robot head with a multimedia touch-screen face, for use on a rescue robot; v1.0 shown on an ATRV-jr base and teleoperated	Joints controllable over a network connection, a tether being acceptable (design criteria) [**?**]; radio link and onboard computer NR	NR	NR
Paripreksya 2.0 [**?**]	Tracked UGV, 4 flippers, 4-DoF manipulator, ROS; teleoperated with a joystick on a host PC	Host PC to Intel NUC over Wi-Fi	NR	NR
TXRob [**?**]	Tracked UGV, camera, microphone, gas sensors; teleoperated	Raspberry Pi 4 on the robot; local Wi-Fi network from a Raspberry Pi control station; PS5 controller	NR	NR
IoT disaster response robot [**?**]	3D-printed UGV, ESP32 and ESP32-CAM, video, GPS, remote lighting; remote-controlled	Bluetooth control (10 m, lab); video over Wi-Fi or cellular (12 m, field)	NR	NR
SERR [**?**]	Mobile robot, Arduino, thermal and ultrasonic sensing; smartphone remote control	Smartphone app, RF module, live video	NR	NR
Rescuer [**?**]	Pioneer-P3DX with Kinect camera; monitored and controlled from a base station	LoRa LPWAN, no infrastructure; 1.6 miles reported; Intel NUC and RoboMaC devices on the robot	Operator-commanded E-Stop bit only; link-loss stop NR	NR

Table 2.1: Representative rescue platforms: platform, control and link/computing against C1-C2 (part a of Table 2.1).

Platform	**Platform and control**	**Link and computing**	**C1**	**C2**
Universal Platform [**?**]	Add-on controller on a DJI Tello drone and a quadruped; operator voice commands, autonomous navigation, person detection	Wi-Fi, cable or cellular (GSM); Raspberry Pi operating the drone; cloud used for voice-command processing only	NR	NR
Active scope camera [**?**]	Tethered continuum robot, 80 distributed sensors incl. microphones and speakers; operator interface	Multiple PCs on a ROS network; optical cable inside the robot	NR	NR
SAR assistant robot [**?**]	Arduino Mega, thermal IR array, ultrasonic, accelerometer; autonomous survivor approach	NR	NR	NR
Amphibious caged UAV [**?**]	Ground-air UAV, separation cage; piloted with a handheld remote controller	Remote-controller link with live imagery; real-time object detection	NR	NR
ERS [**?**]	4-wheeled vehicle that must be driven manually (stated), YOLOv8 victim detection	Onboard Jetson Nano and USB camera; detections uploaded to a Firebase database and shown in a mobile app; control link NR	NR	NR
QUADRESCUE [**?**]	ROS quadruped, 12 servos, RGB-D camera, IMU, GPS; operator-assisted, joystick control	Raspberry Pi 4B with a Raspberry Pi Pico; Wi-Fi to the rescue team; 200 m control range reported	NR	NR
LoRa surveillance robot [**?**]	Two-motor ground vehicle with Arduino Uno and ESP32-CAM; operator keypad; the paper also calls it a drone	LoRa command link; ESP32-CAM video over Wi-Fi; Blynk cloud dashboard. Reported specifications: 15 km open and 5 km urban range, 8 h battery, 1.5 s video latency	Stop is an operator command; link-loss stop NR	NR
This work	4WD UGV, Arduino UNO and Raspberry Pi 4; teleoperated	Local Wi-Fi, all services on the robot, no Internet (Sections 3.2, 3.9)	Firmware dead-man; drive removed within 2042 ms of the last command in 30 of 30 logged trials (Sections 3.5.2, 5.1)	P3 watchdog; service restored in 30 of 30 SIGKILL trials (Sections 3.7, 5.4)

Table 2.2: Representative rescue platforms: C3-C4, code availability, evidence and text read (part b of Table 2.1).

Platform	C3	C4	Code	Evidence and conditions	Text read
Survivor Buddy 1.0/2.0 [**?**, **?**]	Design goal: the screen lets the survivor videoconference with responders or family [**?**]; two-way link not described or evaluated in either paper read	USD 3,600 (v1.0) and USD 1,800 (v2.0), head only, Table II of [**?**]	NR	Design-process case study with a v1.0/v2.0 comparison table [**?**]; quasi-experimental study with four middle-school girls and their friends, not statistically significant [**?**]; two simulated US&R scenario studies [**?**]	Full text of both Murphy papers; abstract of [**?**]
Paripreksya 2.0 [**?**]	NR	NR	NR	Competition: WRS 2020 remote-participation category on a field built by the team, with video of each task submitted	Full text
TXRob [**?**]	One-way: video and microphone audio to operator (stated)	"Low-cost", no figure	NR	Physical prototype; uneven surfaces in a simulated disaster environment	Full text
IoT disaster response robot [**?**]	NR (video to operator only)	"Affordable", no figure	NR	Physical prototype; lab tests and field experiments	Full text
SERR [**?**]	Speaker and microphone provisioned; not yet implemented (stated)	NR	NR (data on request)	Simulation: detector validated in MATLAB	Full text
Rescuer [**?**]	Audio: a low-cost walkie-talkie mounted on the robot as a proof of concept, enabled by the operator through a LoRa bit for push-to-talk; the paper states that the victim could communicate directly with the operator, with no audio quality or delay reported. Video: Kinect footage recorded on the robot, not streamed (stated)	"Low-cost" walkie-talkie only; no total	NR	Simulation (Gazebo with ESP32 modules) and an indoor laboratory test of the robot; the 1.6-mile LoRa distance is stated for the base-station link	Full text
Universal Platform [**?**]	Speech synthesis described as able to give protocol information to rescued individuals or serve as a communication medium; not among the reported tests; victim-to-operator audio NR	NR	NR (data on request)	Physical tests of commercial robots: drone indoors and outdoors (wind 23 km/h, microphone noise 70 to 100 dB), quadruped indoors; voice-command and 3D-reconstruction results reported	Full text
Active scope camera [**?**]	Microphones locate the survivor's voice; speakers stated as used to talk to survivors, and also for sound-based position estimation; the reported search experiment played a simulated victim voice from a loudspeaker; no video to the victim	NR	NR	Physical prototype; simulated rubble at the Fukushima Robot Test Field with a mannequin; sensor-communication trials	Full text
SAR assistant robot [**?**]	NR	"Inexpensive, commonly available parts", no figure	NR	Physical prototype; experiments on sloped terrain	Full text
Amphibious caged UAV [**?**]	NR	NR	NR	Physical prototype; flight and endurance tests	Full text

Table 2.2: Representative rescue platforms: C3-C4, code availability, evidence and text read (part b of Table 2.1).

Platform	C3	C4	Code	Evidence and conditions	Text read
ERS [?]	NR (one-way detection alerts)	"Low-cost" processor, no total	NR (data, code and weights on request)	Dataset: detector fine-tuned and evaluated on 6,837 images (many synthetic or augmented); on-device runtime of 80 to 140 ms per frame stated; field test NR	Full text
QUADRESCUE [?]	NR (operator identifies victims in the visual feed; GPS localization)	"Cost-effective", no figure	NR (code on request)	Physical prototype; field evaluation on difficult surfaces against wheeled platforms; latency and packet loss over distance up to 200 m	Full text
LoRa surveillance robot [?]	NR	NR	NR	Physical prototype; pilot tests described qualitatively; the specifications are stated, and no measurement procedure or data were found in the text	Full text
This work	Two-way WebRTC video and audio, plus text and images (Section 3.9); resource cost in Section 5.5	approx. USD 310, receipt-backed (37,158 BDT; Section 5.9)	Public GitHub repository; no licence file, so no open-source licence has been applied	Indoor laboratory measurements on the robot as built (Section 5)	Own system

The gap that Table 2.1 makes visible is a combination rather than a single missing feature, and none of its parts is claimed as a first. The table shows what each paper reports, which is weaker than showing what each robot lacks. Teleoperation is the mode used in real deployments [?], and the link to the operator fails in the field [?]. Stop categories are standardised [?] and recovery by restart is an established dependability strategy [?], yet, in the text read for Table 2.1 (see its caption), no recent platform reports a stop time bounded independently of its higher-level software (C1) or supervised recovery of its processes (C2). Watchdog timers for drone processors [?] and link-loss failsafes in general-purpose ground-vehicle autopilot software [?] are documented, but no platform in the table reports either for a rescue robot. Two-way communication with the trapped person is a stated requirement [?], a feature of fielded mine-rescue robots [?] and the subject of an HRI literature [?, ?, ?], yet among recent low-cost platforms the victim is mainly a detection target, one planned voice channel is not implemented [?], another is a separate walkie-talkie with no video stream [?], and none reports two-way audio and video with the victim on single-board hardware or its resource cost (C3). Among recent platforms low cost is asserted without a figure (C4). Section 3 brings these established mechanisms together on commodity hardware, and Section 5 reports which properties the current measurements support and which they do not yet cover.